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SHIP AND OCEAN SYSTEMS DESIGN

Ship types

Six StatCode families, and what sets the size of each

Ship and Ocean Systems Design (PNSO)
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In this lecture

1. The StatCode 5 classification: six families of ships, coded in five levels.
2. For each group, the requirement that sets the main dimensions: deadweight, volume, deck area, speed or mission.
3. Where the displacement equation $\Delta = DWT + lightship$ decides the size, and where it is only a check.

TABLE 1 · STATCODE 5 · LEVEL 1

Six Families of Ships, Coded in Five Levels

One ship, five levels

Level 1 · Family
Cargo carrying

Level 2 · Group
Dry cargo/passenger

Level 3 · Sub-group
Container

Level 4 · Ship type
Container ship

Level 5 · Detailed type
Container ship (fully cellular)

Cargo carrying

Bulk carriers · Dry cargo/passenger · Tankers

Our two ships and the reference ship

Work vessel

Fishing · Offshore · Miscellaneous (dredging, research, towing, other activities)

Non merchant

Naval and naval auxiliaries · Yachts · Sail training ships

Non seagoing merchant

Inland waterway cargo ships, tankers, passenger ships, tugs and dredgers

Non propelled

Barges and pontoons

Non ship structure

Platforms, floating docks, buoys, submersibles, hovercraft

TABLE 2 · CARGO CARRYING SHIPS · LEVELS 2–4

Cargo Carrying: Three Groups, Fifteen Sub-groups

Bulk carriers

Bulk dry: bulk carrier, ore carrier

Bulk dry/oil: OBO and ore/oil carriers

Other bulk dry: cement, aggregates and wood-chip carriers

Self discharging: self-unloading bulk carrier

Dry cargo/passenger

Container · our ships: container ship

General cargo · reference ship: general cargo and deck cargo ships

Other dry cargo: heavy-load, livestock and barge carriers

Passenger: cruise and passenger ships

Passenger/general cargo: Ro-Pax ferries

Refrigerated cargo: reefer

Ro-Ro cargo: Ro-Ro, Con-Ro, vehicles carrier

Tankers

Chemical: chemical, chemical/products, edible oil tankers

Liquefied gas: LNG, LPG and CO2 tankers

Oil: crude oil and oil products tankers

Other liquids: water, molasses and juice tankers

TABLE 3 · WORK VESSELS AND OTHER FAMILIES · LEVELS 2–4

Work Vessels, and Ships That Are Not Merchant Ships

Fishing

Fish catching: fishing vessel: trawler, seiner, longliner

Other fishing: fish carrier, fish factory ship, fishing support vessel

Offshore

Offshore supply: platform supply ship, offshore tug/supply ship (AHTS)

Other offshore: drilling ship, FPSO and FSO, pipe layer, offshore support vessel (SOV, DSV, CSV), standby safety vessel

Miscellaneous

Dredging: dredger, hopper dredger

Research: research and survey vessels

Towing/pushing: tug, pusher tug

Other activities: icebreaker, cable layer, crane ship, pilot, patrol, search and rescue, fire fighting, wind turbine vessel, training ship

Non merchant

Naval ships and auxiliaries (frigate to submarine, uncrewed vessels), yachts, sail training ships

Non seagoing merchant

Inland waterway versions of cargo ships, tankers, tugs and dredgers

Non propelled

Barges, pontoons

Non ship structure

Platforms, floating docks, buoys, submersibles, hovercraft

TABLE 4 · WHAT SETS THE SIZE

Five Design Drivers across the StatCode Groups

Size set by	Typical ships (StatCode group)	Role of Δ = DWT + lightship
Deadweight	Bulk carriers, oil tankers (Cargo carrying); hopper dredgers (Work vessel, dredging)	**Sizes the ship:** Δ follows from DWT, then L, B, T
Cargo volume	Container, Ro-Ro, vehicles carriers, reefers, livestock, liquefied gas tankers (Cargo carrying)	Volume fixes L, B, D; the equation gives the draught
People and spaces	Passenger and Ro-Pax ships (Cargo carrying); yachts (Non merchant)	Decks, gross tonnage and stability govern; weight is a check
Deck area and lift	Offshore supply and support, FPSO, crane and wind turbine vessels (Work vessel); heavy-load carriers (Cargo carrying)	Deck area, deck load and stability govern; Δ checks the payload
Power, speed or mission	Tugs, icebreakers, fishing vessels (Work vessel); naval ships (Non merchant); fast craft	**Weight-critical:** lightship grows with power and speed

The exercise on the displacement equation is the first row: a deadweight carrier, where solving Δ = DWT + lightship(Δ) gives the size of the ship.